

Components of the mobile money model

Mobile banking is conducted through a technology platform developed by a mobile financial service (MFS) provider. While it is easily conducted on smartphones, this model has a simple interface that can be used on the most basic cellphone, making it accessible to people living at the base of the pyramid.

Most services can be accessed through SMS, so people are not reliant on an Internet connection. In most cases, no user fee is charged for registration or making deposits, transfers and cashing-out services. Since companies have designed their platforms to work with even the most basic phones, in some cases, voice-based systems are integrated into the service to ensure that those who are not tech-savvy or are under-educated can also benefit.

MFS providers set up and develop technology platforms to enable mobile money services. But, there are a number of critical partnerships required for it to work successfully.

1. If the company is not an MNO itself, then partnering with operators is critical. Most MFS providers partner with several operators to ensure their service is available as widely as possible.
2. The financial regulator of the country where the service is in operation needs to provide a licence to the company to offer the MFS.
3. Businesses that offer banking services need to find banking partners.

Security features

Reduced transaction failure leads to higher customer satisfaction. In the mobile money model, a mobile PIN (mPIN) is required to make money transfers and other functions to reduce the risk of fraud. In cases of theft or loss of smartphone, the subscriber's wallet amount cannot be used without entering mPIN. The application must be compliant with regulatory and industry standards.

Some mobile money applications also include a PCI-compliant platform and triple-DES password encryption with hardware security module key

exchanges. There is also an implementation of internal risk and fraud management suite augmented with external APIs.

Different platforms have different security features and follow different protocols as per the regulation set by their country's government. For example, Alepo mobile money system follows Payment Application Data Security Standard (PA-DSS). Rafael Avigad, vice president - marketing, Alepo Technologies, says, "Blockchain-based ledger improves security by providing a simple auditing mechanism for regulators as well as making the ledger tamper-proof for participating parties (partners, regulators, internal auditors and so on).

Additionally, multi-factor authentication, cryptographically-secure personal data, and tokenisation of payment methods and credentials are used.

Shailendra Naidu Somarouthu, chief executive officer, Obopay, says, "Our processing platform embodies various back-end functions and logics to drive payments. It is based on Java2 Enterprise Edition (J2EE) foundation and supported by a highly-reliable Oracle Real Application Cluster (RAC) database environment. It includes services such as know your customer (KYC), fee engine, velocity limits, reporting and various adapters to financial institutions and payment networks."

Open APIs with high security promise secure encryption to prevent fraud (through hardware security modules), multi-factor authentication and a variety of processes to deliver a PCI/SAS-compliant technology platform that meets the highest standards and best practices for the payments industry.

Aswin Phlaphongphanich, chief executive officer, SawasdeeShop Co. Ltd, Thailand, says, "Since we partnered with leaders from countries like Malaysia, we adopted similar processes and standards in our operations, to ensure the same operating-level procedures. We follow the same IT guidelines for data centres, disaster recovery, network securities and system securities as do all banks in Thailand. This is strictly governed and

monitored by Bank of Thailand under licence regulations."

How mobile money system works

Avigad says, "The mobile money system consists of components that allow interoperability, a unified rules engine that ensures regulatory compliance and helps drive commissions and incentives, and a blockchain-based private ledger that improves auditability and multi-factor engagement in the ecosystem. To fulfill various operations, the mobile money system interfaces with various utility billers, banks, payment gateways and other components of the ecosystem. The customer can use the system for these functions.

"Alepo mobile money, however, is not a cryptocurrency system. It relies on liquidity being maintained by holding the equivalent amount of money as that which is circulating in a third-party account."

Somarouthu adds, "Developers can leverage open APIs to create and deploy Obopay widgets easily. Moreover, Deploy Once, Run Global model offers economies of scale for payment initiatives and a single-core platform for powering carrier money and customised partner deployments in a multi-country, multilingual environment."

Phlaphongphanich explains, "There are three simple steps to complete payment and money transfer in a mobile money system. Simply download the app, complete e-KYC and remit money. On completing the transaction on the app, customers receive the e-invoice via SMS, which helps them track the transaction."

Advantages of mobile money in evolving markets

Most credit card companies charge a 2.5 to 3 per cent transaction fee. Eliminating the card lowers this cost. Mobile money has low transaction costs with improved security. It generates new employment opportunities and creates a platform on which other businesses can grow.

Mobile money services have reduced customers' reliance on cash, thus increasing their financial security